CAPSTONE PROJECT 2

E-COMMERCE AI INTELLIGENCE PLATFORM

Purchase intent prediction • Hybrid product recommendations

ANN

RECOMMENDATION ENGINE

STREAMLIT

RETAILROCKET / PYTHON / TENSORFLOW

ANN results on unseen test data

ROC-AUC	PR-AUC	PRECISION	RECALL
0.8833	0.2254	29.15%	37.12%

F1 SCORE **0.3265**

Decision threshold: 0.94

Because the positive class is rare, PR-AUC and F1 are more informative than accuracy alone.

Hybrid recommendation engine

Behavior signals

View = 1 · Add-to-cart = 3 · Transaction = 5

Co-occurrence

Learn product pairs from training sessions only.

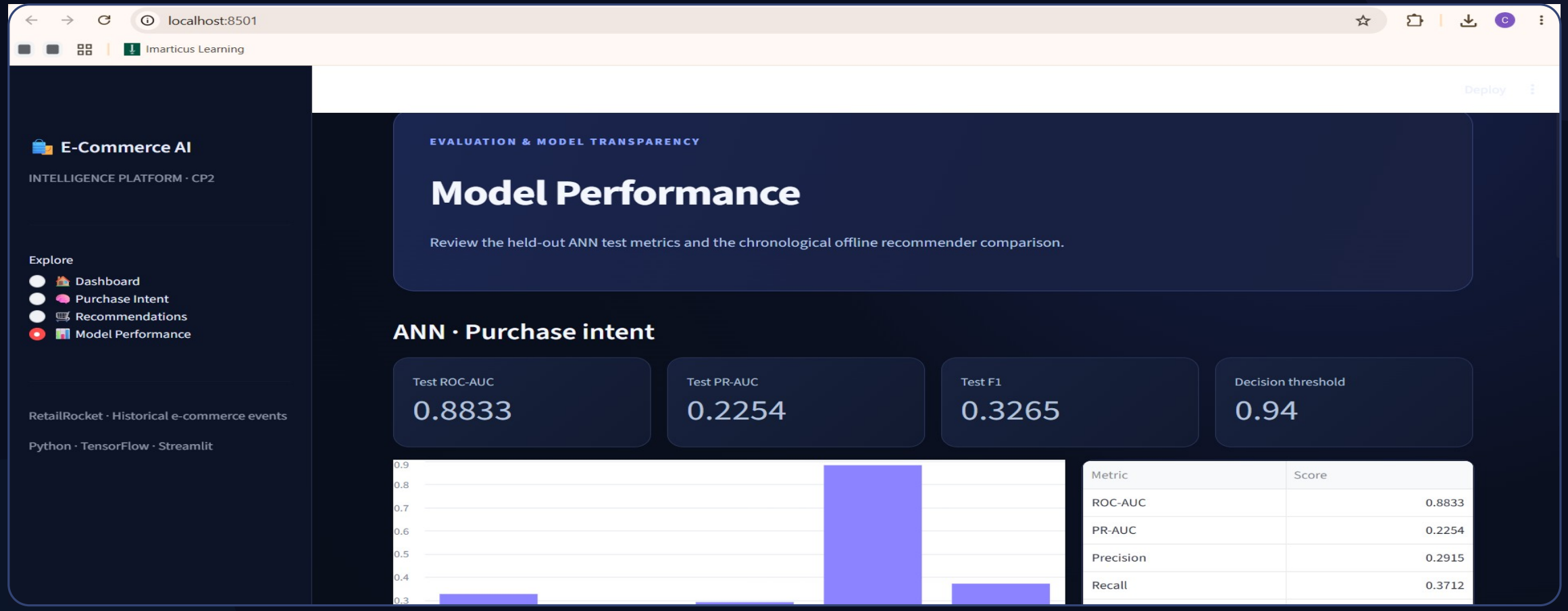
Metadata rerank

Same category ×1.20 · Same parent ×1.10

LIVE RECOMMENDATION

Exclude seen products → filter latest availability → rank Top-N

Live application preview



Dark-themed Streamlit model-performance view